# Component Transfer Can Exceed Full Model Performance: Investigating Post Trained Mixture of Experts

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## Motivation

**SFT+DPO**
OLMoE-0924

**SFT+DPO+RLVR**
OLMoE-0125

post-training

post-training

**Where do post training improvements originates in MoE models?**

Router

Attention

Experts

Most evaluations compare whole models — we go deeper

## Method – Component Transfer

**SFT+DPO checkpoint**
Base model

**Replace with SFT+DPO+RLVR checkpoint**

- ☑ Router
- ☑ Attention
- ☑ Experts (64 MLP networks)

Tulu 3

**Evaluate on tasks & prompting**

| Tasks | Prompting |
|---|---|
| GSM8K | Zero-shot |
| BoolQ | 5-shot |
| ARC-Challenge | Chain-of-Thought |

9 transfer configs × 4 conditions = 36 experiments

## Results

### RESULT 1 — COMPONENT TRANSFER BEATS BOTH SOURCE MODELS

**BoolQ: Attn+Experts (79.2%) > SFT+DPO+RLVR (75.4%) > SFT+DPO (78.0%)**

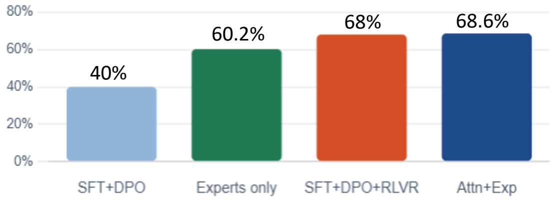
| Model / Config | BoolQ | vs SFT+DPO+RLVR |
|---|---|---|
| SFT+DPO | 78.0 | **+2.6** |
| SFT+DPO+RLVR | 75.4 | — |
| **Attn + Experts ★** | **79.2** | **+3.8** |

★ p<0.001 vs both source models. 95% CI ±1.4%. Full evaluation.

**Key Insight**: Hybrid outperforms both source models, exceeding the performance of full model checkpoints

### RESULT 2 — EXPERT NETWORKS DRIVE REASONING GAINS

**Experts alone recover 72% of the 28-point GSM8K gap**




**Key Insight**: Experts network alone recover 72% of the reasoning gap between models

### RESULT 3 — PROMPTING STRATEGY REVERSES EXPERT EFFECTS

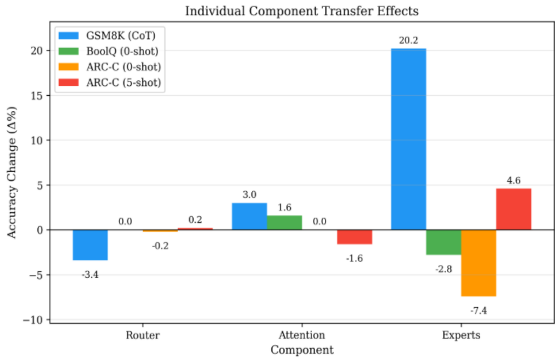
**ARC-Challenge — expert transfer effect**

| Zero-shot | 5-shot |
|---|---|
| Expert transfer | Expert transfer |
| **WORSE** −7.4 pts | **BETTER** +4.6 pts |

**12-point swing from prompting alone**

**Key Insight**: Few shot prompting flips performance losses into significant gains

### RESULT 4 — INDIVIDUAL COMPONENT TRANSFER EFFECTS




**Key Insight**: Attention transfer is a robust default, while expert gains are high-impact but task-specific

## Practical Guidance – How to Merge MoE Models

- **Reasoning tasks**: Use **experts + attention from SFT+DPO +RLVR**.
- **Classification tasks**: Use **experts from SFT+DPO** and **attention from SFT+DPO+RLVR**.
- **Router transfer**: Keep the router paired with its original experts; avoid transferring routers independently.

## Conclusion

- Post-training creates task-specific specialization, not universal improvements.
- Expert networks drive reasoning gains but can hurt performance on simpler tasks.
- Attention mechanisms transfer more consistently across domains.
- Selective component composition can exceed the performance of either source model.
- Future merging methods may benefit from component level operations.

## Paper & Resources